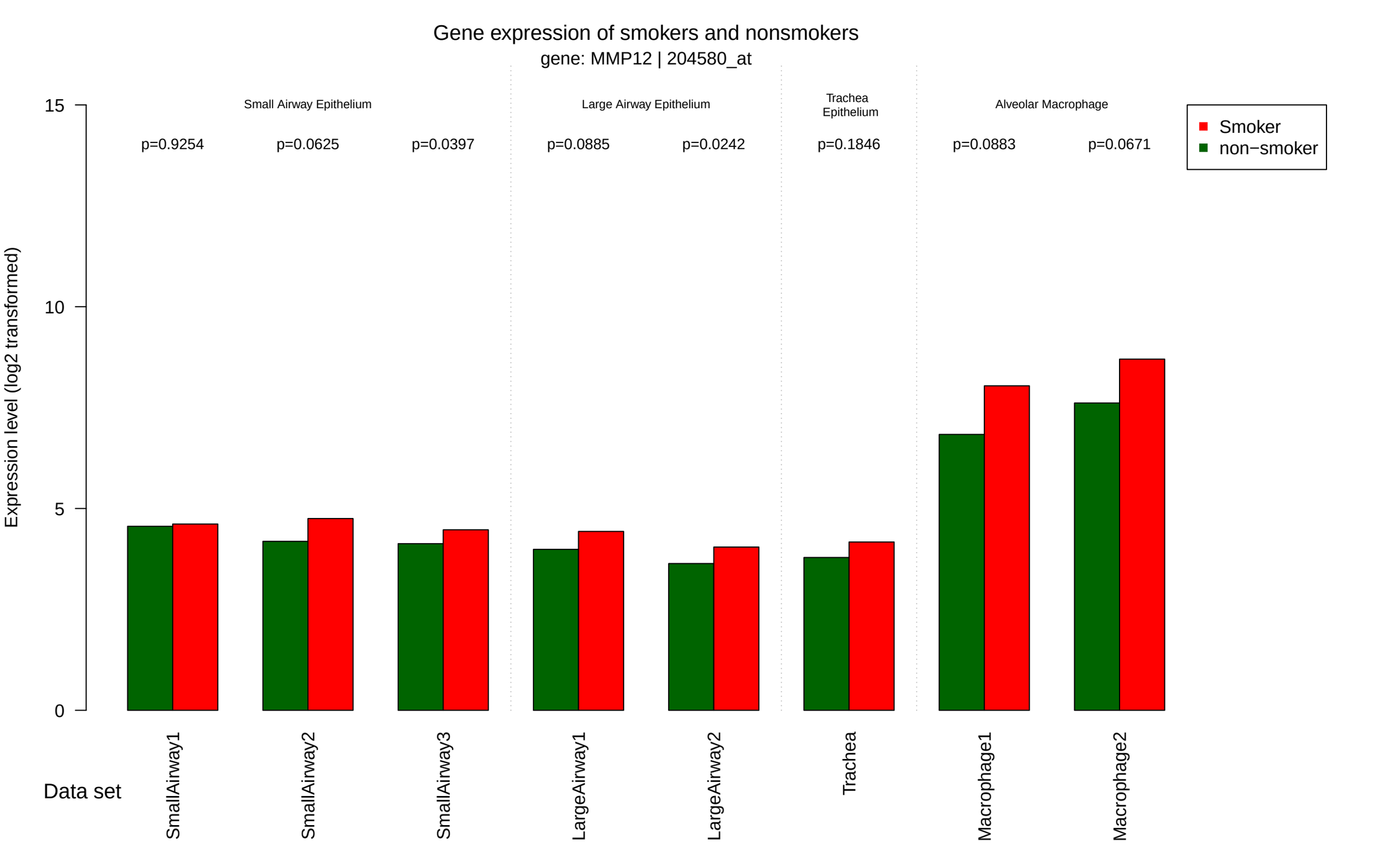
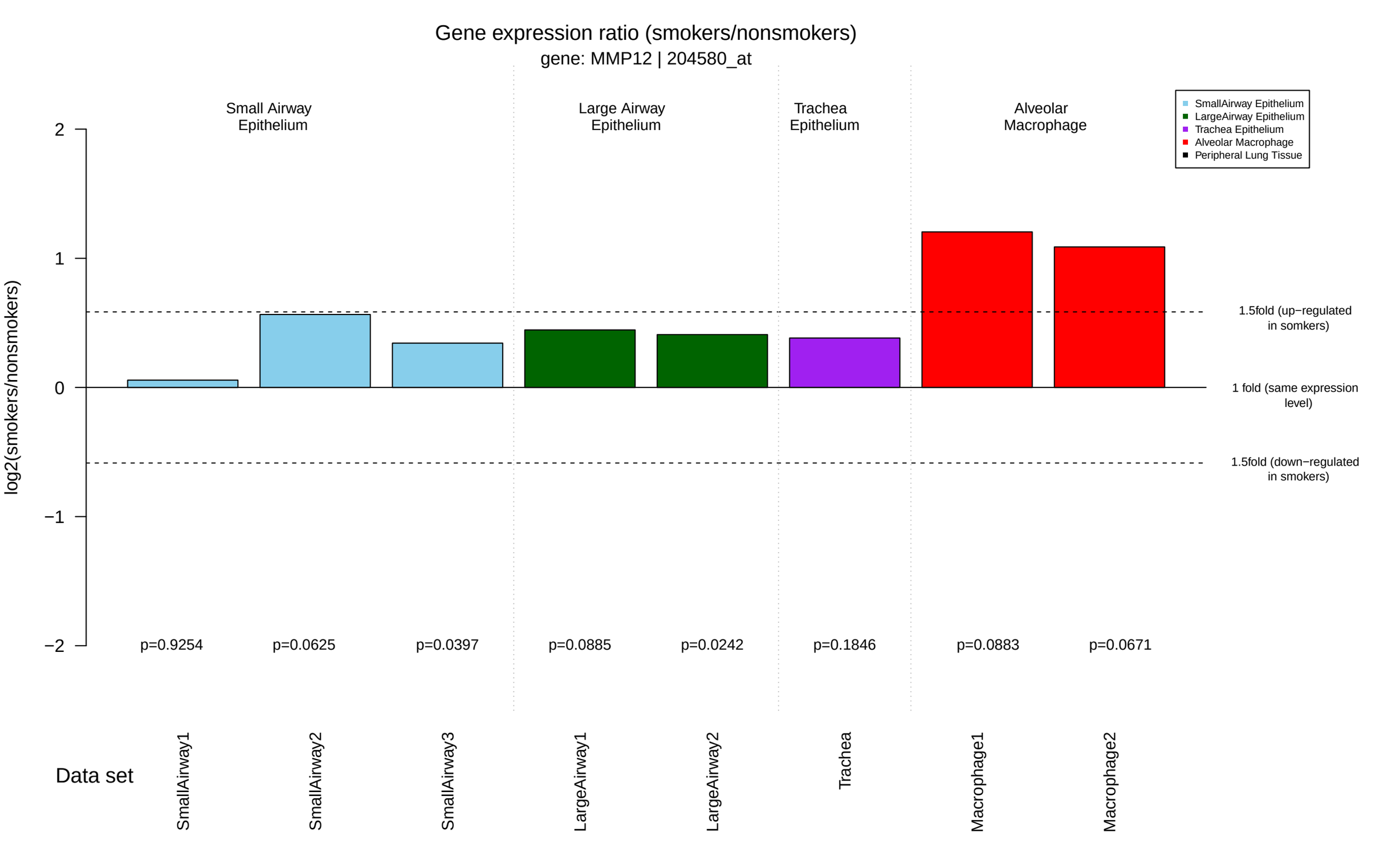
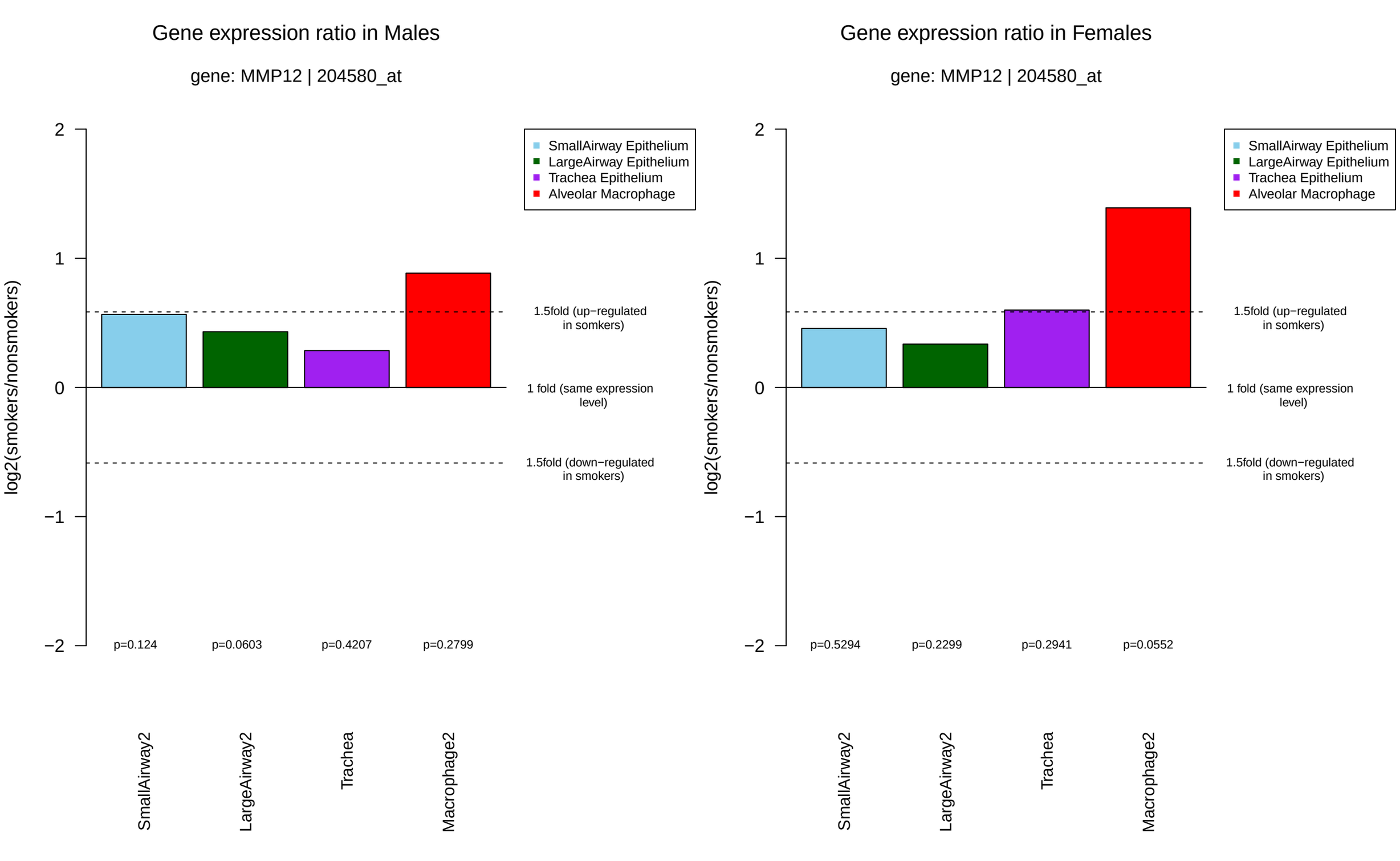
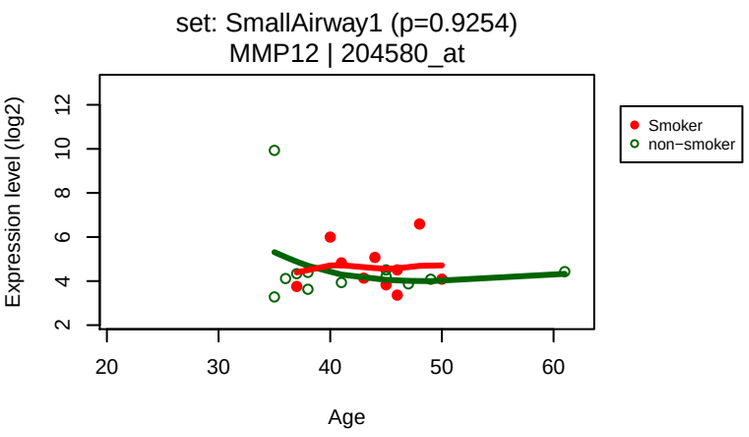
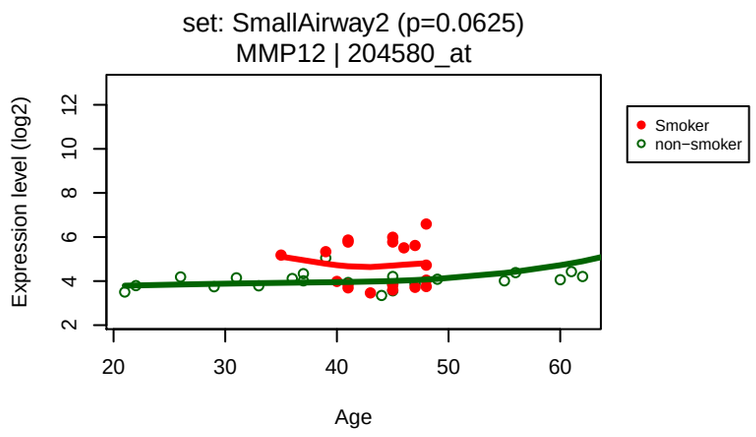
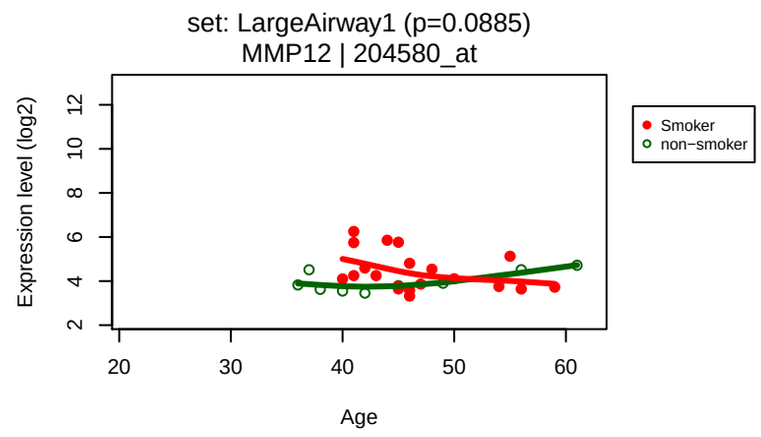

Gene expression of smokers and nonsmokers
gene: MMP12 | 204580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9254
p=0.0625
p=0.0397
p=0.0885
p=0.0242
p=0.1846
p=0.0883
p=0.0671
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

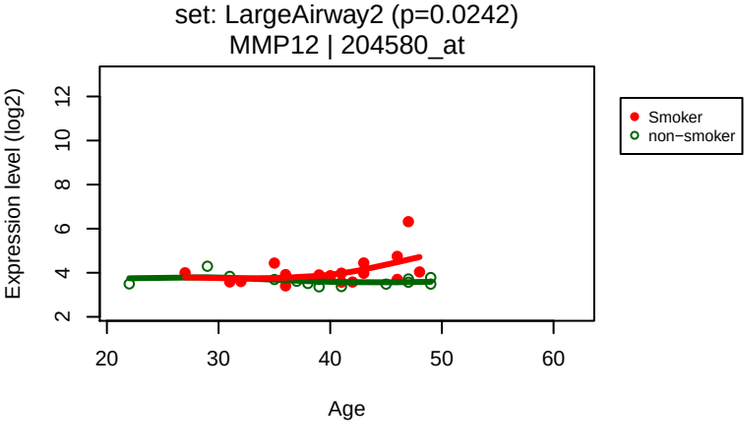

Gene expression ratio (smokers/nonsmokers)
gene: MMP12 | 204580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9254
p=0.0625
p=0.0397
p=0.0885
p=0.0242
p=0.1846
p=0.0883
p=0.0671
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

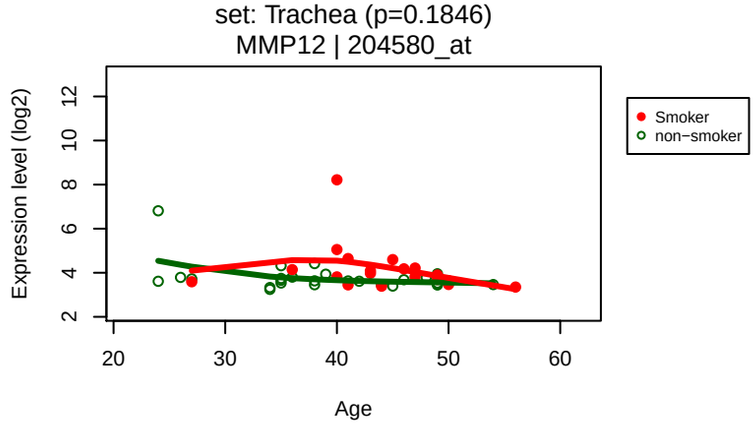

Gene expression ratio in Males
gene: MMP12 | 204580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.124
p=0.0603
p=0.4207
p=0.2799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP12 | 204580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5294
p=0.2299
p=0.2941
p=0.0552
SmallAirway2
LargeAirway2
Trachea
Macrophage2

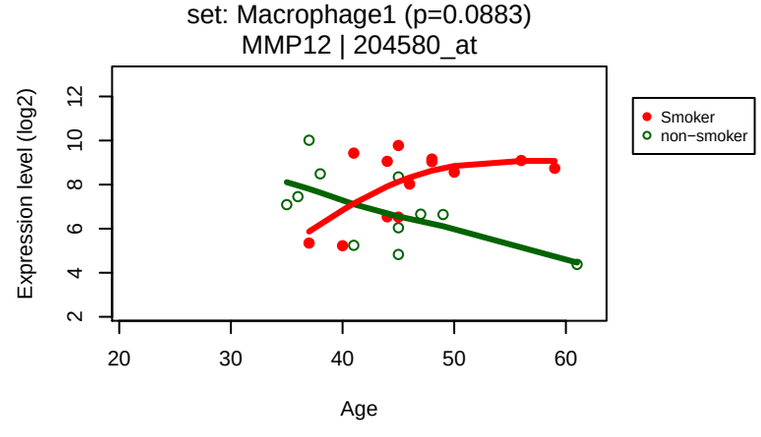

set: SmallAirway1 (p=0.9254)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker

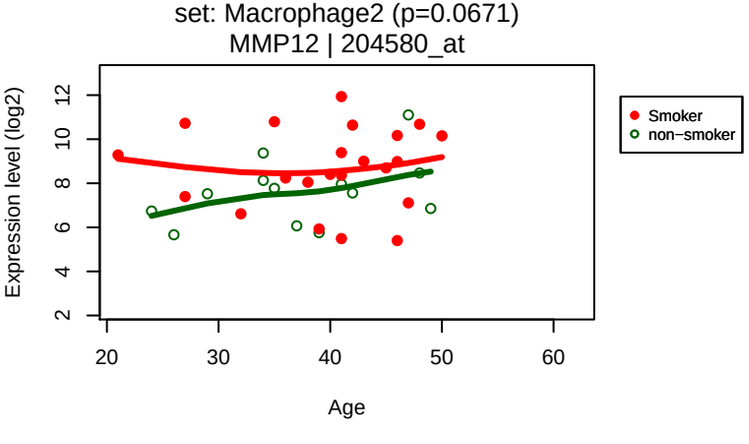

set: SmallAirway2 (p=0.0625)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker

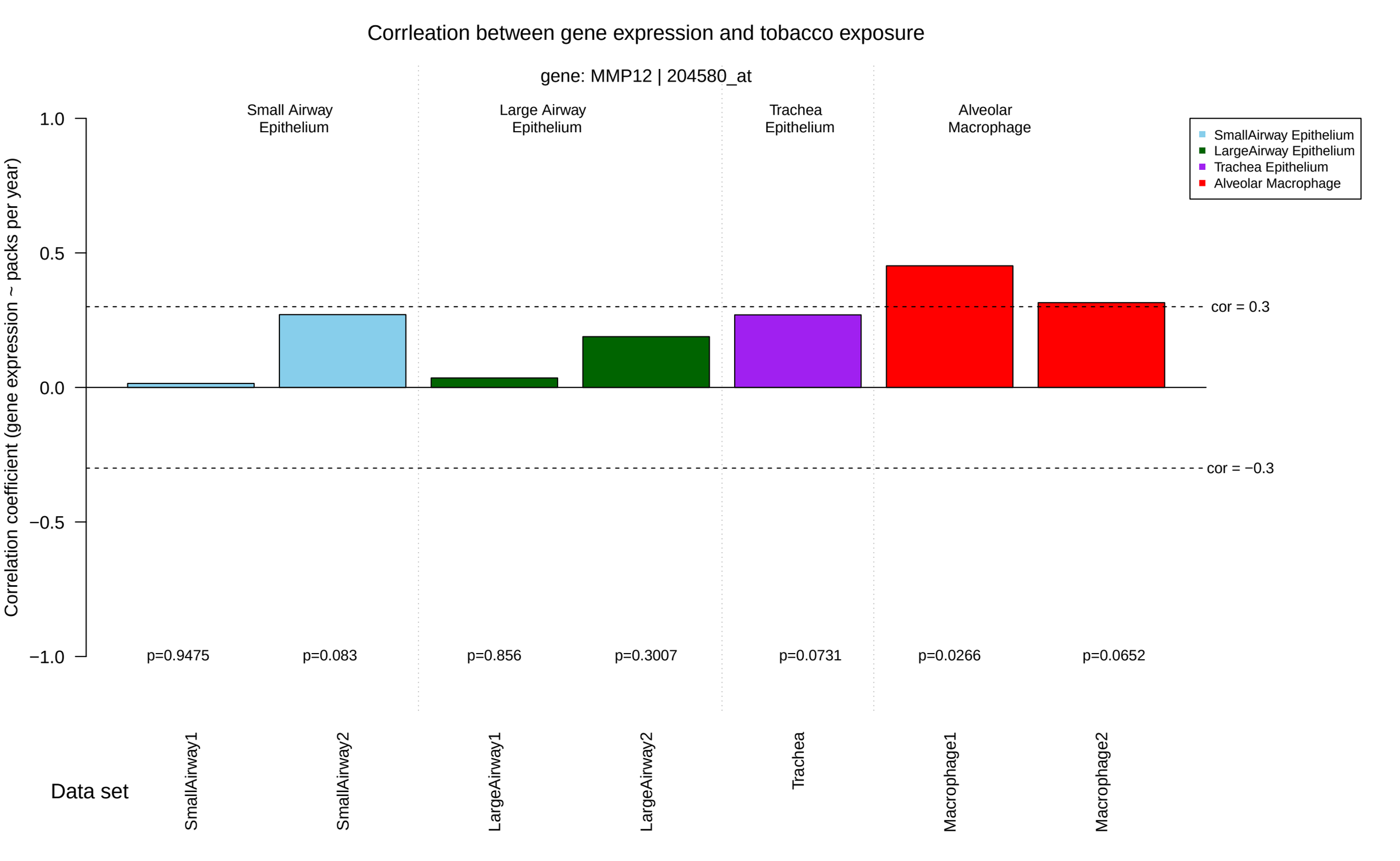

set: LargeAirway1 (p=0.0885)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0242)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1846)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0883)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0671)
MMP12 | 204580_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP12 | 204580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9475
p=0.083
p=0.856
p=0.3007
p=0.0731
p=0.0266
p=0.0652
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2